CHAPTER 6

SHIFTING

"Double-clutch heel-toe" downshifting, the classic racing technique for going down through the gears, has many variations. Upshifting, while simple, requires speed and consistency.

IF RACECARS OPERATED IN a very narrow speed range there would be no need for a gearbox. If the speed varied very little, it would be easy to choose the overall gearing of the car so that at racing speed the engine would always be operating in the RPM range where it makes its best horsepower: what we call its "powerband." In road racing, cornering speeds may vary from 30 m.p.h. to 170 m.p.h., and the gearbox is the device that allows you to operate the motor in its powerband at widely different speeds.

In our experience of teaching over fifteen hundred Three-Day Racing schools over the last 20 years, we have found that the single most frustrating skill for a fledgling racer to learn is proper downshifting.

The main difficulty is that, in racing, the downshift comes at a time when the car (and driver) is busy slowing for an upcoming corner. It's easy for one activity to impose on the level of concentration required for another.

DOWNSHIFTING

The downshift in racing is not as simple as it is on the street: clutch in, shift, clutch out. To illustrate the differences, let's start with the case of a single downshift in a street sedan. Refer to Fig. 6-1, a cross-section of the drive line of the car.

The engine (A) spins the flywheel/clutch assembly (B). When the clutch is engaged—with your foot *off* the pedal—the engine is directly connected to the input shaft of the transmission (C). When the transmission is in gear, a pair of gears are working together—one on the input shaft and one on the output shaft—so that for a particular RPM of the input shaft a different RPM comes off the output shaft side (D). In most transmissions this difference is true for all the gears but top gear. Top, whether it's fourth in a four-speed or fifth in a five-speed, is often a 1 to 1 ratio—that is, for every revolution of the input shaft the output shaft turns once. In all of the lower gears, the input shaft would turn faster than the output shaft. The lower the gear, the faster the input shaft spins relative to the output shaft.

The differential changes the direction of the rotation to one you can use to propel the car forward. At the same time, it changes the RPM of the wheel and tire relative to the RPM of the output shaft. Normally, the ratio of this change is between 3 to 1 and 5 to 1. If the differential ratio were 3 to 1, the wheel would revolve once for every three revolutions of the output shaft.

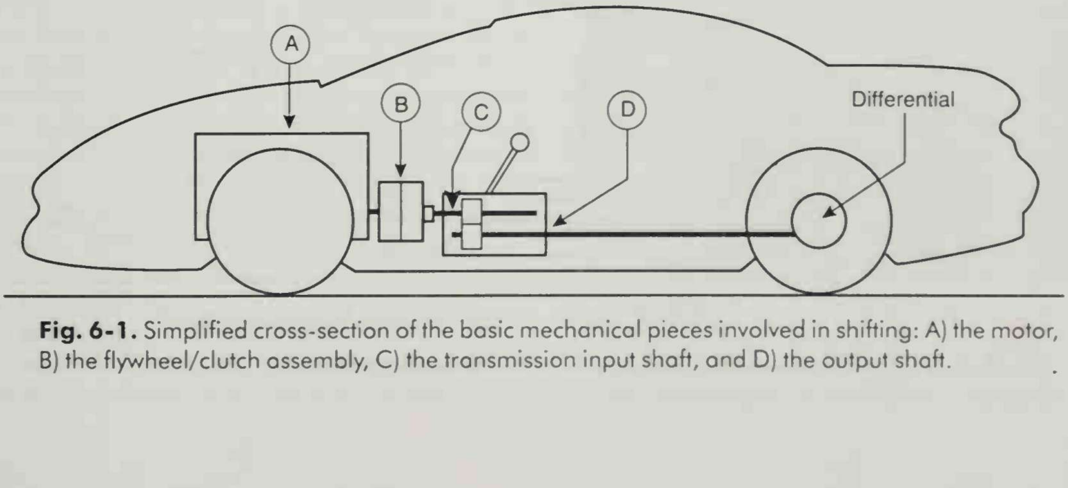

Fig. 6-1. Simplified cross-section of the basic mechanical pieces involved in shifting: A) the motor, B) the flywheel/clutch assembly, C) the transmission input shaft, and D) the output shaft.

RPM Differences

Let's take a look at Fig. 6-2, which shows the RPM of all of these bits when you're attempting a downshift. At 60 m.p.h. the average tire is revolving at 900 RPM. At a 3 to 1 differential ratio the output shaft of the transmission would be spinning around at 2700 RPM. If you were in top gear at 60 m.p.h. in a car with a typical transmission (1 to 1 top gear ratio), the input shaft (C) would be turning around at 2700 RPM as well.

Okay, now you're going to slow the car to 50 m.p.h. to negotiate the upcoming curve. Before you get to the corner you're going to downshift to third gear. You take your foot off the throttle and squeeze on the brake, slowing the car to 50 m.p.h. As you can see in Fig. 6-3, the rear wheels are turning over at 744 RPM, the output shaft is spinning at 2232 RPM, and you put in the clutch. When you do this, the engine is no longer connected to the input shaft—the clutch has broken this connection. With your foot off the throttle and the engine disconnected from the transmission (thereby disconnected from the rear wheels), the engine falls to idle speed—say, 1000 RPM.

Now you move the gear lever out of fourth gear toward third. In this lower gear, the input shaft, when you're done with the shift, is going to be revolving faster than the output shaft. The increase in RPM differs from transmission to transmission, but it's in the range of 25% faster than it would be spinning when fourth is engaged. In order to get third gear engaged, the input shaft will have to spin up to 2790 RPM at 50 m.p.h.: that's over 500 RPM faster than it was spinning when you put the clutch in. In a street car, synchronizers in the transmission speed up the lower gear so that when the teeth on one come into contact with the teeth on the other, they are going at the right relative speed so that there isn't a god-awful *graaaunchh* when the gear lever gets shoved into the third gear.

The bad news is that the transmissions in most racecars *don't have any synchronizers*. You'll have to face the music on this one soon, but for now, let's finish this street car downshift.

In Fig. 6-3, you muscled the transmission into third gear; now the input shaft is turning at 2790 RPM and the output shaft is at 2232 RPM—the appropriate RPM for your 50 m.p.h. road speed. The engine is still at idle: 1000 RPM. If you engaged the clutch like a steel trap—*boom*—it would instantly slow the input shaft to 1000 RPM which, in third gear, translates to only 18 m.p.h. of rear wheel speed.

Since you're going 50 m.p.h. while the rear wheels are only turning over at 18 m.p.h.—that's like locking up the handbrake at 50 (see Fig. 6-4). Fortunately, the

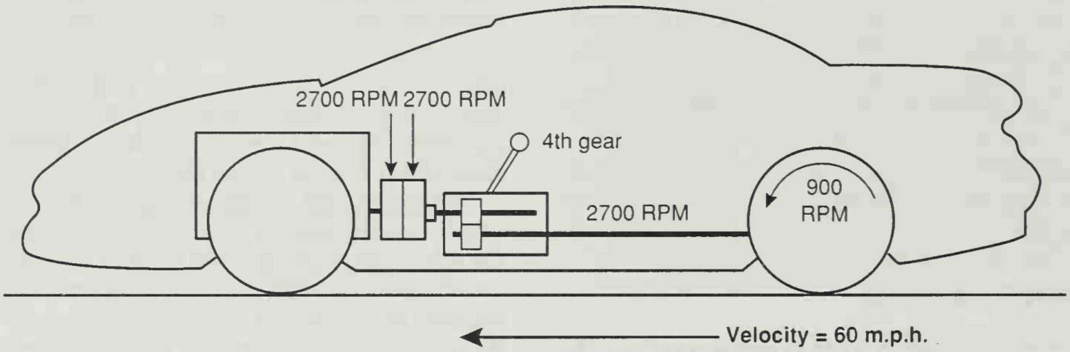

Fig. 6-2. At 60 m.p.h. in top gear the relative RPM of the rear wheels, the output and input shafts, and the engine are displayed above.

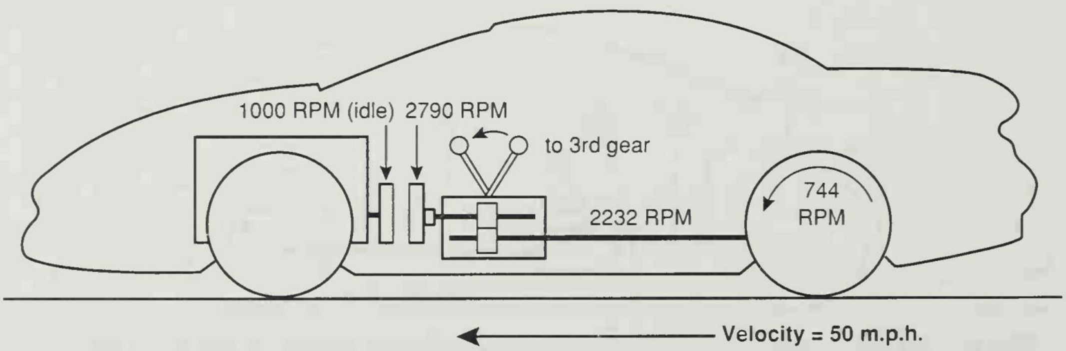

Fig. 6-3. Shifting from 4th to 3rd at 50 m.p.h., synchronizers speed up the input shaft of the transmission to 2790 RPM: the appropriate speed for 50 m.p.h.

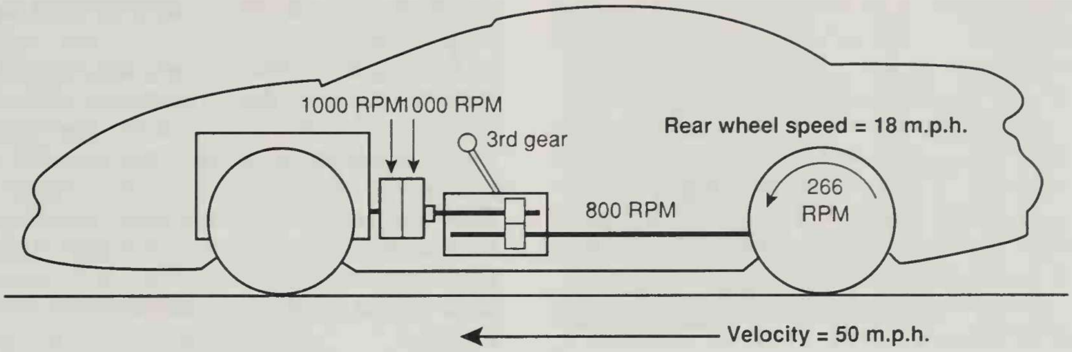

Fig. 6-4. If the clutch were engaged with the engine at idle speed, this converts to just 18 m.p.h. of road speed at the rear wheels. At 50 m.p.h. this is quite a shock as the rear tires have to speed up the engine from 1000 RPM to 2790 RPM.

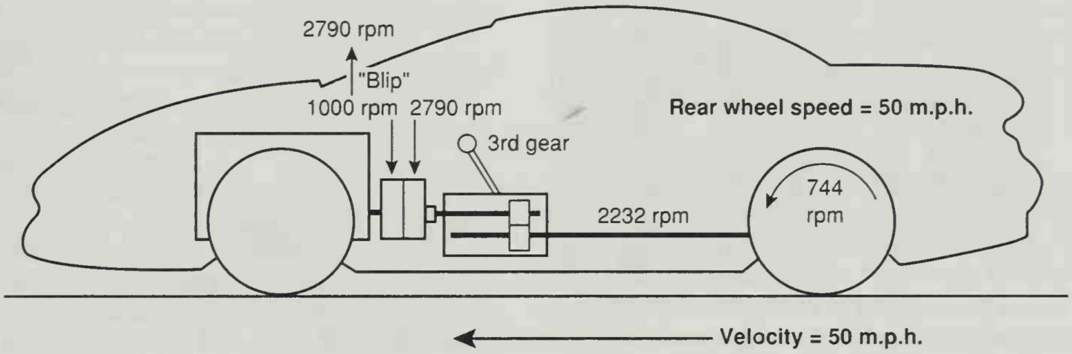

Fig. 6-5. If you "blip" the throttle, bringing the engine's RPM up to about 2790, *then* engage the clutch, the shift will be smooth and not overtax the rear tires' grip on the road.

clutch will slip when you engage it so that the shock to the driveline is minimized somewhat, but just dumping the clutch after doing a downshift is brutal on the balance of the car.

The Blip

Properly done (see Fig. 6-5), you should use the throttle to raise the engine RPM to about 2790 before the clutch is engaged again. You don't squeeze on the throttle and hold it; you tap it. You give it a quick, sharp burst. "Blip" is the perfect word for it. The blip should take place just before re-engaging the clutch so that there is no shock to the drive line when going from a higher gear to a lower gear. On the street this is done to keep the ride smooth. On the racetrack where you're operating close to the tires' limit of adhesion, a downshift without a blip can use up *all* the traction of the rear tires and spin the car in the straight-line braking zone. Even for a car with a synchronized transmission, a blip before clutch release is mandatory.

Heel-and-Toe

Since your left foot is on the clutch, and your right foot is on the brake pedal, you have to make some adjustments in order to blip the throttle. The way to do it is to adjust the pedals in the car so that with the ball of your foot pressing on the brake pedal there's a few inches of right foot left over to roll your foot to the right and tap the throttle, blipping the engine up the required RPM so that when the clutch comes out everything will be smooth and gentle.

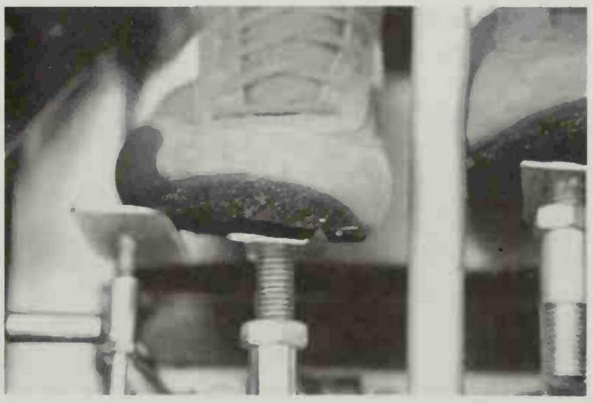

Fig. 6-6. A proper brake/throttle relationship locates the throttle close by and just slightly below the brake pedal when the brakes are on hard.

This downshifting method is called "heel-and-toeing," although, as we've just described it, it doesn't involve your heel or your toe. The phrase was coined more than forty years ago, when many racecars had the pedals arranged so that the brake pedal was on the right, the clutch pedal on the left, and the throttle was between them and about six inches lower. Under braking, the driver had the ball of his right foot on the right-hand pedal and when the "blip" was needed, pushed down on the throttle with his heel—hence, heel-and-toe. Pedal arrangements have changed but the term lingers on.

The pedals in most street cars aren't set up to facilitate heel-and-toeing, so drivers are forced to go through some real contortions to be able to touch the brakes and throttle at the same time. To do it right you have to do more than just be able to reach both pedals with the same foot. You need to accurately control how hard you're pushing on the brake pedal, a skill that's lost to many drivers who twist their feet into deformed postures to accommodate the awful pedal positioning in their cars.

What Downshifting Is Really For

We ask this basic question of every racing school class. The most frequent (and incorrect) answer is, "to help slow the car down." In a racecar with good, durable brakes (the majority of modern racecars), downshifting to help the car slow down is unnecessary. The *brakes* slow the car down. You downshift to get the car in the proper gear to exit the corner.

There *are* cars with marginal braking systems, most notably showroom stock racers which have brakes designed for the demands of the street environment. Their brakes, especially in an endurance event, may fade away to nothing, at which time you use whatever you can to slow the car for the corners. In this case you certainly *do* use the downshifting to slow the car down—but it's a last resort.

Choosing Your Gear

You decide which gear is the proper gear for a corner simply by picking one and trying it. If the engine is turning over too slowly—that is, it's operating in an RPM range below where it makes its best power—the gear you've chosen is too high. Next time, try a lower gear.

At the other extreme, if the engine hits its redline RPM before the exit of the corner, the gear you've chosen is probably too low. Try the next highest gear next time. When in doubt, use the higher gear since it's less likely to damage the motor, and in the future you'll be more likely to take the corner faster than slower.

Where To Downshift

Rather than drawing an X on a track map, we try to help the driver make the decision based on avoiding shifting too soon or too late. Take a look at Fig. 6-7 to see the range of choices.

For simplicity's sake, let's say that this is a third gear corner, approached in fourth gear. You certainly can't downshift the instant you go to the brake pedal. At this speed the car would be going too fast for the engine to be under the redline RPM in the next lowest gear. You need to let the car lose enough speed so that when the clutch comes out in the next lowest gear you don't over-rev the motor. In this case let's say that the earliest point at which you can downshift comes at point A.

If the throttle application point in this corner is at point B, that would be the latest possible point for the downshift: shifting anywhere between point A and point B is OK—a wider range of choices than you might have guessed.

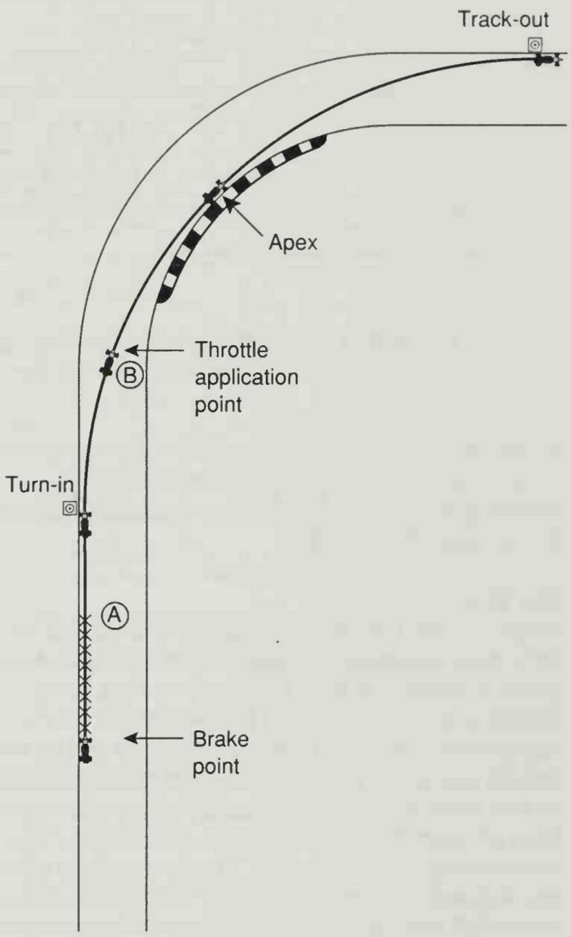

Fig. 6-7. There is an earliest and latest point at which the downshift can be done. Anywhere between point A and point B is fine, keeping in mind that, given the choice, it's less risky to downshift while the car is going straight.

There is one other potential restriction. If you have the choice, downshift while the car is going straight rather than when the car is turning. If you blow it and chirp the rear tires when the clutch comes out in the lower gear, the car is much more likely to spin if it's under cornering load than if it's not. There will be times when you're forced to downshift while the car is turning, and you'll have to do it well, but given the choice, you take less risk by downshifting on the straight.

Double-Clutch Downshifting

In shifting gearboxes with synchronizers, the function of the blip is to smooth out the clutch engagement after moving to the next lowest gear. Since most racing transmissions do not have synchronizers, the blip not only cushions the clutch engagement, it also allows you to synchronize the speeds of the input and output shafts. Here's how it's done.

With your foot on the brake you push in the clutch and shift to neutral from a higher gear. You let the clutch out, and then, still in neutral, you blip the throttle.

Since, with the clutch engaged, the motor is directly connected to the input shaft, your blip easily spins up the shaft to the higher RPM. The instant after you blip, push in the clutch and move the gear lever into the third gear position. Once it's there, let the clutch back out again. This last bit has to happen fast before the extra RPM created by the blip falls away.

By performing this series of maneuvers, you've synchronized the gears, going from fourth to third without grinding. The secondary benefit is that since the clutch re-engaged in third gear with the engine turning over at the right RPM for the road speed, you shifted into third gear smoothly without lurching off the drive-line.

Use Figs. 6-8 through 6-18 as a visual guide:

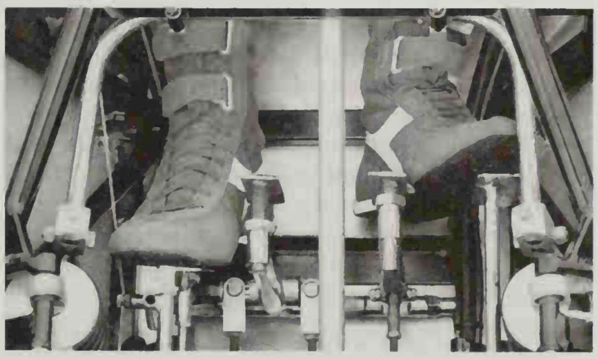

Fig. 6-8. Full throttle on approach to corner.

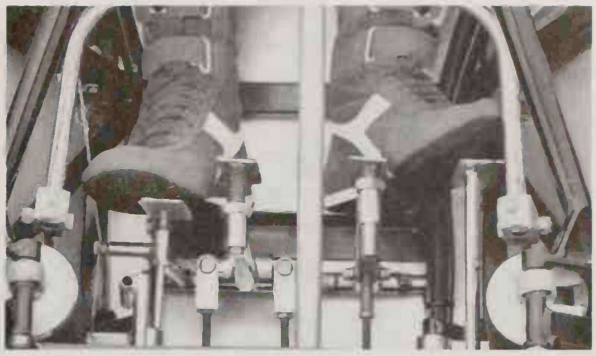

Fig. 6-9. Come off the throttle.

Fig. 6-10. Step onto the brakes.

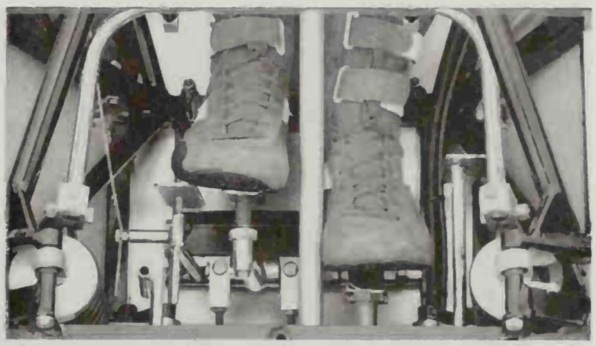

Fig. 6-11. As the car decelerates, push in the clutch.

Fig. 6-12. Move the gear lever down to neutral.

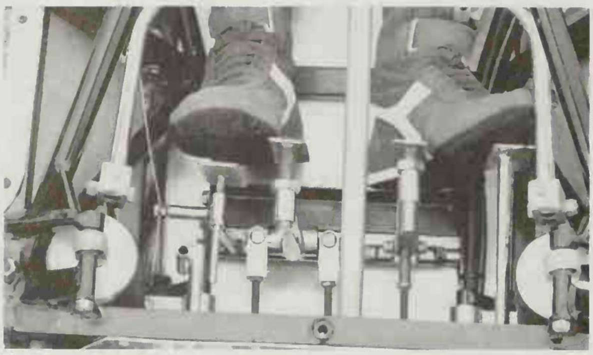

Fig. 6-13. Then let the clutch back out.

Fig. 6-14. Blip the throttle.

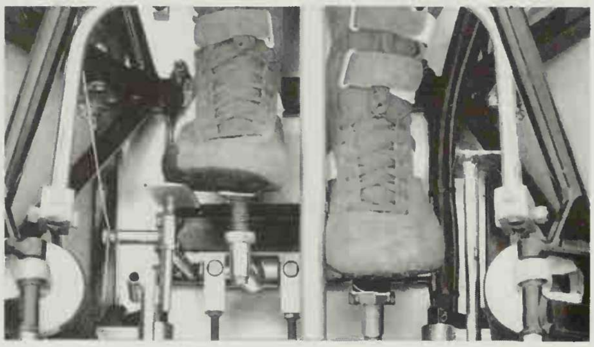

Fig. 6-15. Push in the clutch pedal.

Fig. 6-16. Move the gear lever into the lower gear.

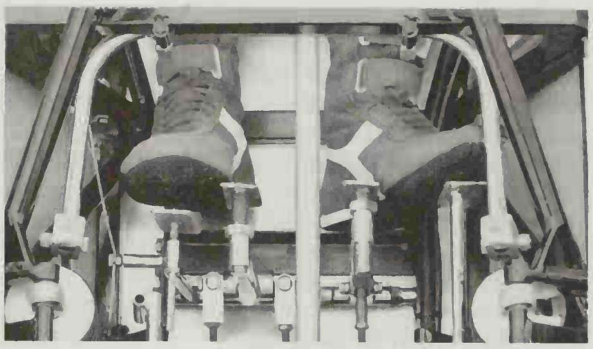

Fig. 6-17. Let clutch out, turn in, then relax brake pedal pressure.

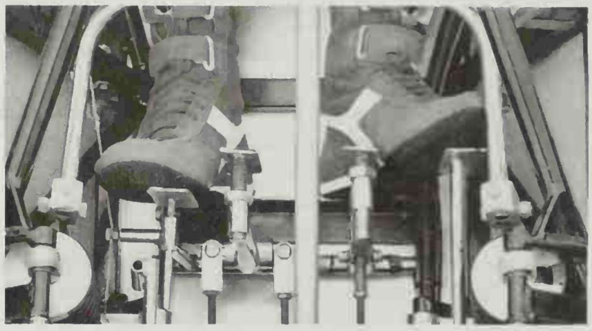

Fig. 6-18. Apply throttle through apex and away from corner.

Be Clear About The Routine

When you first try to double clutch, it seems like your hands and feet just won't work together. It's very easy to get flustered, especially if, like most of our Three-Day Racing School Students, you're sitting in a real race car for the first time.

It helps to think of the process as doing two normal shifts to accomplish one—a shift to neutral, and a shift to the lower gear. Think about it. The shift to neutral is like any other shift—clutch in, move the gear lever, clutch out. The next half of the double-clutch is simply another normal shift—clutch in, move the gear lever to the next lowest gear, clutch out.

The only difference is that when you stop in neutral you give the motor (and the input shaft) a blip.

Dorsey Schroeder invented the downshift jingle which he'd perform at every drivers school he would teach, and he swears it helps. Using the same meter as your basic samba dance, it goes:

Verse 1,4th–3rd
clutch in, neutral, clutch out
blip
clutch in, next gear, clutch out
Verse 2, 3rd–2nd
clutch in, neutral, clutch out
blip
clutch in, next gear, clutch out
Verse 3, 2nd–1st
clutch in, neutral, clutch out

blip
clutch in, next gear, clutch out
If he'd had his way, he'd have everybody in a samba line getting the jingle down before trying it in the cars.

> *"Double clutching is important to me— it's my balance. I always do it, and always use the clutch. I've had times when the clutch broke— a pushrod or a hydraulic line—and I'd still use the clutch pedal for a timing deal. I've learned not to release brake pressure when blipping the throttle and if I change my routine I get segmented and throw the timing off. It just didn't work right without double clutching."*
>
> *—Dorsey Schroeder*

MISTAKES

Keeping the Clutch Dipped

The most common error when trying to learn to double clutch downshift is forgetting to let the clutch back out when you stop in neutral and give the motor a blip. When the engine isn't directly connected to the input shaft of the transmission, which would be the case if you let the clutch out, the blip isn't as effective at speeding up the input shaft. There is some drag going on between the motor and the input shaft, and a blip, even with the clutch depressed, will speed up the shaft some. The blip is much more effective, however, if you positively connect the engine and input shaft by letting the clutch back out in neutral.

Slow Move to the Next Gear

Once the throttle is blipped and the input shaft spun up to speed, the shift to the next lowest gear has to be done right away. If you wait too long, the motor and the shaft slow down and the synchronization is lost, resulting in a more difficult (and louder!) downshift. If for some reason the lever doesn't go into the next lowest gear following the blip in neutral, you need to blip the throttle again before trying again.

Trying to Shift Too Fast

Just because you're driving a racecar does *not* mean that every motion you make behind the wheel has to be fast. Most downshifts come at a time when you're decelerating the car for an upcoming corner and there is more time available to do it than you might think. You don't have to perform like "The Flash," especially if double clutching and heel-and-toe is new to you. Avoid trying to move the gear lever from the higher gear to the lower gear in one motion, hoping that the clutch will come out and the blip will occur at the precise instant the lever passes through neutral. Move the lever twice: once to neutral and then again to the next lowest gear. You can take as much time as you like doing the first half of the downshift—clutch in, lever to neutral, clutch out. But, as mentioned above, once you blip, you need to complete the shift quickly.

GETTING GOOD

There's no secret to developing the ability to downshift smoothly, instinctively. You need to do three million heel-toe double clutch downshifts. Well, maybe not that many. If you drove an hour a day, every day of the year, and did three downshifts for every minute on the road you'd do 60,000 a year. That would do it.

If you're serious about it, get a standard transmission street car and set up the pedals so that you can heel-and-toe it. From that day on, *always* heel-and-toe double clutch downshift. In a year, you'll never have to consciously think of downshifting again.

To Clutch, or not to Clutch

A lot of people ask us if top-level professionals go through every step of this downshifting process. The answer is that some do, some don't. The biggest variable is when and if the driver uses the clutch.

> *"I never double clutched in my life. I always blip on the downshift and use the clutch to go down, even with the sequential gearboxes, but I don't necessarily let the clutch out in neutral. I've always been very gentle with gearboxes and when I taught racing with Skip's, it was the one thing I didn't do that we taught. Guess I just give bigger blips than the average driver."*
>
> *—Danny Sullivan*
>
> *Author's Note: We never could get Danny to buy into double clutching, but the single clutch works if you blip higher than you would if you engage the clutch in neutral. Blipping higher spins the input shaft up the same amount, even without the direct connection between engine and input shaft that engaging the clutch in neutral provides.*

At the very beginning of the process, when you lift off the throttle and go to the brake pedal, there is a moment when the tension on the gears is released as the car goes from driving the rear wheels with the motor to the motor being forced to turn by the rear wheels. In this little transition period, you can easily pop the car out of the higher gear into neutral without using the clutch and without any strain on.the gear box. Now you're all set up to do the blip without having used the clutch. The second phase of the shift would be to push in the clutch, shift to the lower gear, and let the clutch out—unless of course you choose to do this part of the downshift without the clutch too. If the blip was perfect, not too high or too low, the gears should be perfectly synchronized and the gear lever should just fall into the next lowest gear without any grinding from a mismatch.

Many experienced drivers who are very good at matching the input and output shaft revs, shift without the clutch as a routine. Blipping the throttle as they pause in neutral allows them to do this without harming the transmission.

If, however, your technique is less than perfect, using the clutch takes the shock out of the inevitable mismatches in gear speeds that occur when you blip too much or too little. Using the clutch provides a little slip in the system and a minor mismatch just wears a little more off the clutch plate—a much less expensive part than gears and dog rings.

"I have used the clutch on upshifts and downshifts all the way up to the point where we started using sequential gearboxes in the Indy cars. I used the standard double clutch downshift technique I learned in my Three Day School all the way through, F/Ford, Barber/Saab, Indy Lights—everything. Not using the clutch risks hurting the car. You've got to have some mechanical sympathy."

—Bryan Herta

A lot of inexperienced drivers rush into trying to shift without the clutch, mostly to be like the big guys. We think it makes much more sense to learn how to do the double clutch heel-toe downshift using the clutch and leave it up to the individual to decide when they, (and their transmission maintenance budget), are ready to move on to clutchless shifting.

Keep in mind that in downshifting there is no lap time advantage in shifting without the clutch. Downshifts come when the car is slowing down and it's the brakes that determine how quickly the car covers this section of the course. Getting the car into the lower gear a tenth of a second faster than the competition doesn't necessarily mean that the braking process is going to take a tenth of a second less. In most cases, there's more than enough time to get the downshift done, regardless of the method you use. One thing is for sure: you want to end up at the end of the race with a gearbox that works as well as it did when the race began. Grinding gears on every shift uses up the gearbox and results in defeat by your own hands—and feet.

Inside a Racing Gearbox (Or: Why Bother to Double Clutch)

A common explanantion for matching the speeds of the input and output shafts of the transmission is that when you change gears you're forcing a gear on the input side to come into contact with the gear on the output side. This is not true.

Although the above makes for a more vivid mental picture, the gears, or more properly, the *pairs* of gears, are constantly intermeshed with each other. Moving the gear lever selects which pair of these intermeshed gears you choose to use. When the gear lever is in neutral, none of these pairs of gears are hooked up to the input shaft of the transmission.

The gear lever moves into place a device called a dog ring. The dog ring spins at the same speed as the input shaft, and when the gear level is in neutral they spin freely, out of contact with any pair of gears.

When you move the gear lever into the first gear position, that slides the input-shaft dog ring along until its teeth (called dogs), engage the dogs on the pair of gears that make up first gear. At this point, power that is delivered from the engine through the clutch to the input shaft is transferred to the input shaft half of first gear, to the output shaft, and from there through the differential to the driving wheels.

All of your double clutching and blipping is an effort to make the dog ring on the input shaft spin at the same RPM as the gear into which you plan to shift. When it all goes wrong, those horrible noises you hear coming from inside the gearbox are from the dogs on the dog ring clattering into the dogs on the gear you're aiming for.

Skipping Gears

You can either go down through the gears one at a time, or you can go from fourth to first rather than stopping at third and second along the way, but like everything else, there are pros and cons for each method of downshifting.

The defense for skipping gears is the argument that doing one shift is less distracting to the braking process than three. In our data collection we have found that even very experienced and talented drivers tend to release some brake pressure when they blip the throttle for downshifts. Three downshifts would mean three variations in brake pedal pressure, so it stands to reason that you'd want to do just one to minimize braking changes.

The difficulty with skipping gears comes in timing the shift. If you drive the same racecar a lot, you get accustomed to the amount of time you need to wait in order for the car to slow down enough so that engaging the next lowest gear will not over-rev the motor. It's a matter of internal timing. You don't consciously count "one thousand one, one thousand two," but you develop a sense for when the clutch can come out in the next lowest gear.

If you decide to skip gears and go directly from fourth to first you have to remember to triple the amount of time it takes before you can let the clutch out at the end of the downshift. Yet in the heat of battle, you're apt to revert to the single gear timing for the clutch release, which would do serious over-rev damage to the motor. If you're confident you can keep the timing right, there's no disadvantage to skipping gears other than the fact that it isn't exactly the subconscious routine you'd like your downshifting to be.

In some cases you're forced to skip gears in downshifting. Some cars slow down so fast that you simply can't move the gear lever fast enough to go through every gear. A Formula Atlantic car, for example, can lose over 60 m.p.h. per second under braking. You might choose to go fifth to third to first instead of directly to first, but stopping at each gear along the way is out of the question.

"I skip gears downshifting in situations where it's, like, a fifth gear straight and a second gear corner. I skip because I find the braking more stable that way. Going through every gear, if you're perfect with the blips, the car isn't going to notice but, in practice that's difficult. Skipping keeps the car more stable for me."

—Jeremy Dale

Shifting to Help Braking

In most entry-level racecars, the brakes are durable enough to slow the car repeatedly without much help from engine braking. In cars with marginal brakes, this engine braking may contribute a critical amount of braking help; and choosing to skip gears on the approach to corners may end up overtaxing the braking system. it's not uncommon in racing to lose some or all of the braking ability of the car—the only way you can continue in the race is by using the engine and gearbox to get the car slowed down for the corners. In this case you back off early (sometimes even shutting off the ignition), let the car slow to a speed that won't over-rev the motor in the next lowest gear, and shift it down. It's not an ideal situation but, especially in endurance racing, many of the finishers are the walking wounded, and the ethic is to get the car to the finish no matter what it takes.

UPSHIFTING

The shift up to a higher gear in either a racing gearbox or a stock transmission is the same—clutch in, move the lever up a gear, and clutch out. There is absolutely no need for a blip in the middle of the shift. In the upshift situation the input shaft needs to slow down somewhat in order to synchronize with the next highest gear ratio and this happens naturally as you back off the throttle and depress the clutch. Blipping between upshifts only makes the mismatch worse.

Unlike drag racing, you don't power shift—that is, keep your accelerator foot to the floor while you operate the gear lever and clutch. You should lift off the accelerator, taking the strain off the gear set, and dip the clutch quickly as you move the lever up. You don't necessarily have to push the clutch pedal all the way to the floorboard. You can feel the release point just past the end of the freeplay in the clutch pedal. An inch or so of this hard pressure area is sufficient to create enough slip in the drivetrain to reduce the mechanical shock of the upshift.

"I don't always get the clutch all the way down to the floor—sometimes you only move it halfway, but you get a little bit of disengagement and it makes the gear change that much softer on the gearbox."

—Bryan Herta

Clutchless Upshifts

Some drivers argue for upshifting without using the clutch at all. Again, it's a matter of how good you are at matching the speeds of the input and output shafts. With a very close ratio transmission, just the act of backing off the throttle between gears slows the input shaft enough that the gears end up perfectly synchronized and the gear lever falls into the next highest gear. An expert driver will develop a sense of timing of how long to pause before going back to full throttle to accommodate the synchronization. But, if you don't get the timing right you can really damage the transmission.

We feel that you should use the clutch on upshifts. We've tried it both ways and found that there is no time saved by upshifting without using the clutch. It takes a good driver around .2 seconds to go from full throttle to zero throttle and back again to full during an upshift. If your right foot can move the distance from full throttle to off and back again in two tenths, your left foot can surely do the same thing with the clutch. The benefit is that the clutch cushions the impact of dog against dog and the shock of sudden full power on the driveline. If it were faster, it might be worth the risk, but in test after test we can find no lap-time advantage in not using the clutch on upshifts.

> *"Everybody seems to want to speed shift. It's not any faster. I can tell you because we measured it on the computer."*
>
> *—Bryan Herta*

Bigger Cars

As you move up into faster and more powerful cars, it is possible that you will be faced with the situation where the clutch is the weak link in the driveline—the piece that will wear out first. Each shift and each slip of the clutch plate wears it out a little bit. With big horsepower and a lot of grip, the clutch may wear out before the end of the race due to this slippage, so you may be forced by circumstance to avoid using the clutch for upshifts. In this case, you'd better have the synchronization right or you'll save the clutch at the expense of the gearbox.

> *"I always upshift without the clutch, because I find it smoother. I think it also saves the clutch, which isn't the weak link in a lot of cars, but can be in something like a GTS car. I figure if you never push the clutch going up through the gears, you're probably not going to wear it out or break it."*
>
> *—Jeremy Dale*

Upshifting Speed

It *is* important to develop the ability to upshift quickly. Any time spent between gears is time where the car is no longer accelerating. Minimizing the time spent between gears lengthens the acceleration time, yielding higher straightaway speeds and lower elapsed time down the straights. But here's the catch: the more you rush it, the greater the chance of a missed shift. It takes anywhere from 3/4 seconds to 1-1/2 seconds to recover from a missed shift, and, if the motor blows as a result, it can take even longer.

One of the problems is that as you try to do it faster, you tend to tense up, squeezing the gear lever with a death grip that causes you to lose all sensitivity of the lever's location. You don't have to use more *force* to move the lever faster. You can still hold the gear lever with your thumb and two fingertips: just increase the hand speed (or wrist speed in a formula car). All in all it's the old risk vs. reward trade-off. There are tenths of seconds of lap time in fast shifting, so you try to develop the skill and avoid the pitfalls.

Shift Point

The normal routine when accelerating up through the gears down the straights is to keep your right foot hard on the floor until the tachometer reaches the engine's redline, then shift it up. Much has been written about determining shift points, but taking the motor up to the engine builder's or team manager's recommended redline before shifting pretty much covers it. In powerful cars, especially in the lower gears, the engine is accelerating so quickly that you may need to anticipate the redline by several hundred RPM in order to keep from overshooting.

Rev Limits

There are very few constraints in driving racecars. You're pretty much free to make independent decisions about most driving-related issues. A major constraint is that you have to drive quickly without abusing the motor by taking it over its RPM limit. This constraint is there all the time, whether you're racing your own car or driving for a team.

There is nothing artificial about the redline. Some engines are relatively durable: you can over-rev them from time to time and they don't blow up immediately. But there are some that are real hand-grenades, and one missed shift creates an oily, expensive mess. Shifting is the biggest danger area for motors.

Automatic Shifting

If the current trend toward automatic and semi-automatic gearboxes continues, most of this chapter becomes moot. The computer and the engine management system together select the right gear for the situation. But before you get to Formula 1, you'll have to spend years dealing with non-synchro transmissions. Becoming a good shifter, even though it's not the most fun part of driving racecars, is indispensable in learning to be a talented racer. It's a skill worthy of your attention and effort.